Code Visualization

Divyansh Goel 2024201

Instructions to run

Please install these libraries depending on your system install commands, especially idx2numpy

numpy idx2numpy sklearn seaborn matplotlib pandas

I have written all the code in the main.py, so simply run it using
python main.py
I have also shown the TSNE plots in the main.py

Code Explanation

Step 1 Input
I inputed the images and labels for both train and test for only 0, 1 and 2 digits. Then generated 100 random samples to make the final train and test numpy arrays.

Step 2 Normalize
The train and test samples were in the form of 28x28 arrays, so I converted them to 784x1 so that we have a single column vector with all the features.
The image pixel values are between 0 to 255, so to normalize and make all values between 0 to 1, I divided by 255

Step 3 MLE
I applied the formula for mean and variance as taught in class-

Covariance is $\frac{1}{N}\sum_i (x_i - \mu)(x_i - \mu)^\top$ $\quad \hat{\mu}_{MLE} = \frac{1}{n}\sum_{j=1}^{n} x_i$

By applying these formulae I got the mean related to each label and also the combined mean. Similarly, the covariance matrix for each label and combined covariance matrix.

Since the inverse did not exist (I tried that out and every time inverse did not exist because we have small values in the matrix and finding determinant by multiplying so many small values is like something very small which is taken as 0 det), I added a small kI to each matrix where I is the identity matrix, so that the inverse can exist like taught in class.

Step 4 Discriminant Analysis
I made 2 functions for lda and qda which use the formula of discriminant functions and return the maximum discriminant function value for a given x. These formulas are also from class-
For lda

$$(*) \quad P(x|\omega_i) \sim N(\mu_i, \Sigma)$$

$$g_i(x) = (\Sigma^{-1}\mu_i)^T x - \frac{1}{2}\mu_i^T \Sigma^{-1}\mu_i + \ln P(\omega_i)$$

For qda

$$\text{Case III} \quad P(x|\omega_i) \sim N(\mu_i, \Sigma_i) \rightarrow \text{quadratic discriminant.}$$

$$g_i(x) = -\frac{1}{2}\ln|\Sigma_i| - \frac{1}{2}\left[(x-\mu_i)^T \Sigma_i^{-1}(x-\mu_i)\right] + \ln P(\omega_i)$$

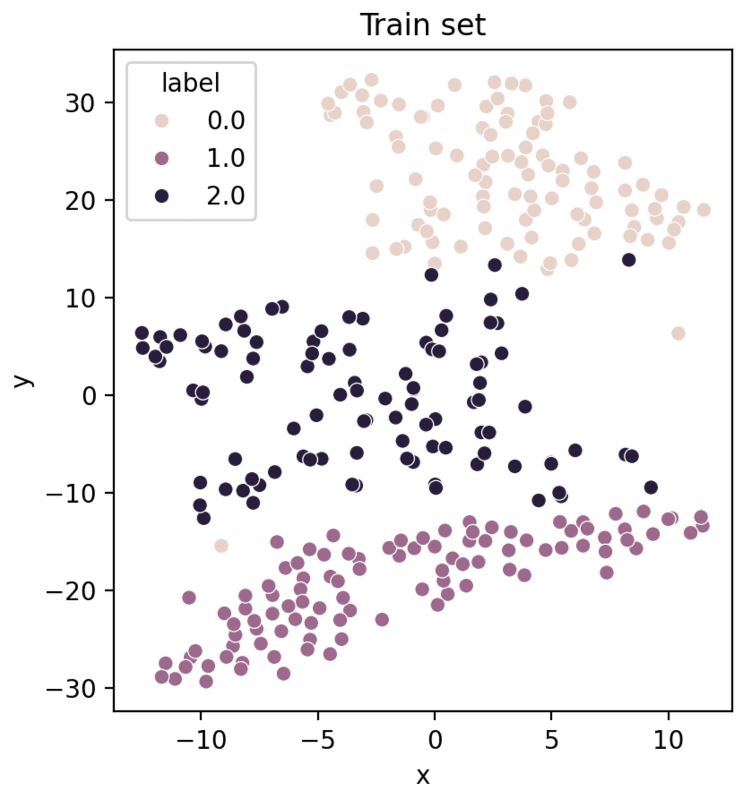
Using these formulae we get the decision for a particular x and I simply ran it on all the test dataset. The priors are equal in our case so they don't matter in the calculation. I reported the accuracy of lda and qda using the formula

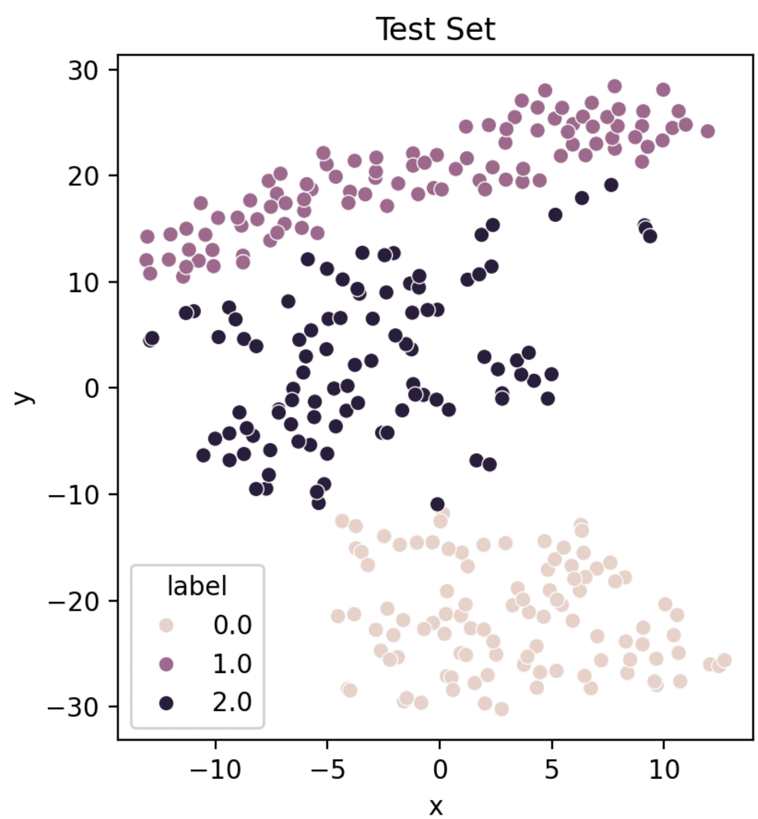
$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

I also showed the discriminant function values for some sample test points.

TSNE Plots

I plotted the TSNE plot individually on the train set which has 300 images and similarly the test set which also has 300 images. In both of them the classification in clusters based on different digits are kind of separated. So we have clusters based on digits 0 1 or 2.


Train set
label
0.0
1.0
2.0
30
20
10
0
−10
−20
−30
y
−10
−5
0
5
10
x


Test Set
label
0.0
1.0
2.0
30
20
10
0
−10
−20
−30
y
−10
−5
0
5
10
x

Sample Output

LDA Accuracy for 0 = 0.92
LDA Accuracy for 1 = 0.99
LDA Accuracy for 2 = 0.8
QDA Accuracy for 0 = 1.0
QDA Accuracy for 1 = 0.99
QDA Accuracy for 2 = 0.99
LDA Accuracy = 0.9033333333333333
QDA Accuracy = 0.9933333333333333
QDA performed better
Discriminant values of random test point from 0 index 91
Prediction, g0x, g1x, g2x - (0, 251.40141927832022, 246.98842296889882, 245.00666451199436)
Prediction, g0x, g1x, g2x - (0, -373.3027917342438, -24482.430716851, -7103.105114961014)
Discriminant values of random test point from 1, index 19
Prediction, g0x, g1x, g2x - (1, 34.4252597841172, 37.220047059714815, 35.02487319681511)
Prediction, g0x, g1x, g2x - (1, -6021.428941762993, 2180.814373148887, -2557.6005626370384)
Discriminant values of random test point from 2, index 4
Prediction, g0x, g1x, g2x - (2, 143.91154310132683, 143.11821950422552, 144.62892148309214)
Prediction, g0x, g1x, g2x - (2, -7363.160033569991, -11788.327871961556, -1259.6147718265188)